

Intermediate Macro

Chapter 18: The Goods Market in an Open Economy

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18.5 Saving, Investment, and the Current Account Balance



- Chapter 18 introduces an **open-economy model of goods market equilibrium** by adding net exports to the demand for domestic goods.
- A real depreciation will improve the trade balance when the proportional increase in relative quantities (the sum of the proportional increase in exports and the proportional decrease in imports) exceeds the proportional real depreciation. This condition is called the **Marshall-Lerner** condition.

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- Chapter 18 considers the short-run goods market in isolation from financial markets, so we assume that the interest rate is fixed and that the real exchange rate is a policy variable.
- In keeping with the analysis of Chapter 19, a more precise way to state these assumptions is that the home and foreign price levels are fixed and the nominal exchange rate is a policy variable.
- Since the price levels are fixed, the nominal exchange rate determines the real exchange rate.
- In addition, production is assumed to respond one-for-one to changes in demand without changes in price, so demand determines output.

The Demand for Domestic Goods

18.1 The IS Relation in the Open Economy

- In an open economy, the **demand for domestic goods**, Z is given by

$$Z = \underbrace{C + I + G}_{\text{domestic demand for goods}} + X - \frac{IM}{\varepsilon} \quad (18.1)$$

- Note that foreign goods are different from domestic goods, so we cannot just subtract the quantity of imports IM .
- We must express the value of imports in terms of domestic goods.
- Recall that ε is the real exchange rate, defined as the price of domestic goods in terms of foreign goods.
- Equivalently, $1/\varepsilon$ is the price of foreign goods in terms of domestic goods.

The Determinants of Imports

18.1 The IS Relation in the Open Economy

- Imports are the part of domestic demand that falls on foreign goods.
- What do they depend on?
- They clearly depend on domestic income. Higher domestic income leads to a higher domestic demand for all goods, both domestic and foreign.
- They also clearly depend on the real exchange rate. The more expensive domestic goods are relative to foreign goods, the higher is the domestic demand for foreign goods.
- Thus, we write imports as:

$$IM = IM(Y, \varepsilon) \quad (18.2)$$

(+,+)

The Determinants of Exports

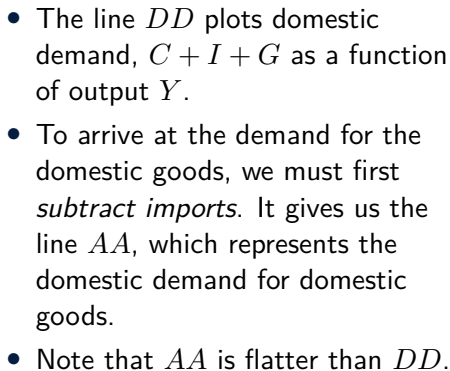
18.1 The IS Relation in the Open Economy

- Exports are the part of foreign demand that falls on domestic goods.
- What do they depend on?
- They depend on foreign income: Higher foreign income means higher foreign demand for all goods, foreign and domestic, so leads to higher US exports.
- Exports also depend on the real exchange rate. The higher the price of domestic goods in terms of foreign goods, the lower the foreign demand for domestic goods.
- We can write exports as

$$X = X(Y^*, \varepsilon) \quad (18.3)$$

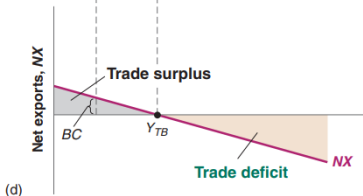
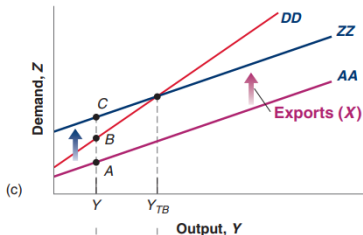
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18.1 The IS Relation in the Open Economy



The Demand for Domestic Goods and Net Exports

18.1 The IS Relation in the Open Economy



- Finally, we must *add exports*. It gives us the line ZZ , which represents the demand for domestic goods.
- We can also characterize the behavior of net exports as a function of output. This is represented as the line NX .

Equilibrium Condition in the Open Economy

18.2 Equilibrium Output and the Trade Balance

- Equilibrium in the goods market requires that the demand for domestic goods equals the production of domestic goods, namely that

$$Y = Z$$

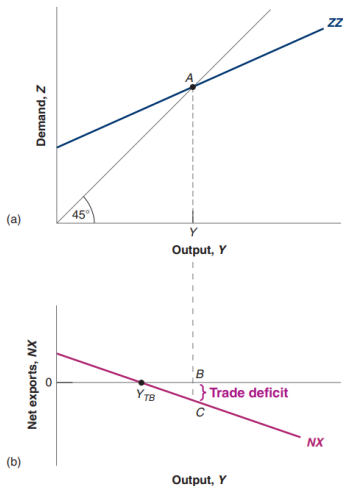
- Substituting equation (18.2) and (18.3) into the demand for domestic goods results in a new IS relation:

$$Y = C(Y - T) + I(Y, r) + G + X(Y^*, \varepsilon) - \frac{IM(Y, \varepsilon)}{\varepsilon} \quad (18.4)$$

Equilibrium Output and Net Exports

18.2 Equilibrium Output and the Trade Balance

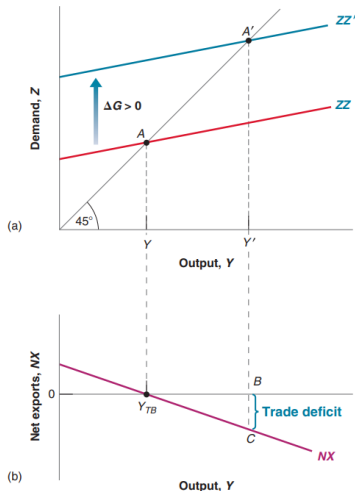
- In general, equilibrium does not require balanced trade.



Increases in Domestic Demand

18.3 Increases in Demand—Domestic or Foreign

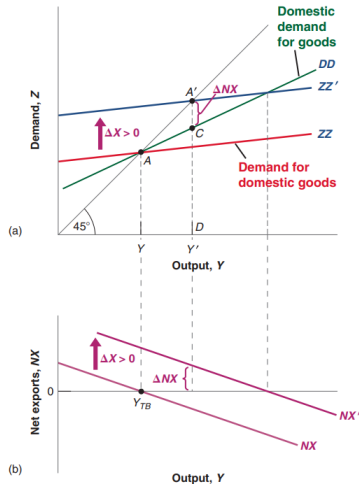
- Suppose the economy is in a recession and the government decides to increase government spending in order to increase domestic demand and, in turn, output.
- What will be the effects on output and on the trade balance?



Increases in Foreign Demand

18.3 Increases in Demand—Domestic or Foreign

- Consider now an increase in foreign output Y^* . This could be due to an increase in foreign government spending G^* .
- The ZZ and NX curves shift up by the same amount.



Fiscal Policy Revisited

18.3 Increases in Demand—Domestic or Foreign

- We have shown that shocks to demand in one country affect other countries.
- These interactions also complicate the task of policymakers, especially in the case of fiscal policy.
- Governments typically do not like to run trade deficits, because deficits require borrowing from the rest of the world.
- In the absence of coordinated action, an expansionary policy by an individual country in the midst of a worldwide recession will likely generate a trade deficit, because the increase in income will increase imports.
- Coordinated expansions may be difficult to arrange.

The G20 and the 2009 Fiscal Stimulus

18.3 Increases in Demand—Domestic or Foreign

- The **G20** was created in 1999 and is now a group of leaders from 20 advanced and emerging countries.
- In November 2008, the G20 leaders met to coordinate their response to the existing crisis in terms of both monetary and fiscal policies.
- Over the next few months, most countries indeed adopted discretionary measure, aimed at either increasing private or public spending.

Depreciation and the Trade Balance: The Marshall-Lerner Condition

18.4 Depreciation, the Trade Balance, and Output

- Recall the definition of net exports:

$$NX = X(Y^*, \varepsilon) - \frac{IM(Y, \varepsilon)}{\varepsilon}$$

- A real depreciation has two effects:
 - ① a quantity effect (an increase in exports and a reduction in imports), which tends to increase the trade balance, and
 - ② a price effect (an increase in the relative price of imports), which tends to reduce the trade balance.
- The net effect will be positive if the quantity effect is greater than the price effect, a condition known as the **Marshall-Lerner condition**.

Derivation of the Marshall-Lerner Condition (1)

18.4 Depreciation, the Trade Balance, and Output

- Start from the definition of net exports

$$NX = X - \frac{IM}{\varepsilon}$$

- Assume trade to be initially balanced, so that $NX = 0$ and $X = IM/\varepsilon$ or, equivalently, $\varepsilon X = IM$.
- Recall that Marshall-Lerner condition is the condition under which a real depreciation, a decrease in ε , leads to an increase in net exports.
- To derive this condition, first multiply both sides of the equation by ε to get

$$\varepsilon NX = \varepsilon X - IM \quad (18A.1)$$

Derivation of the Marshall-Lerner Condition (2)

18.4 Depreciation, the Trade Balance, and Output

- Now consider a change in the real exchange rate of $\Delta\varepsilon$.
- The effect of this change on the left side of (18A.1) is given by $(\Delta\varepsilon)NX + \varepsilon(\Delta NX)$.
- Note that, if trade is initially balanced, $NX = 0$, so the first term in the expression is equal to zero and the effect of the change on the left side is simply given by $\varepsilon(\Delta NX)$.
- The effect of the change in the real exchange rate on the right side of (18A.1) is given by $(\Delta\varepsilon)X + \varepsilon(\Delta X) - (\Delta IM)$.
- Putting the two sides together gives

$$\varepsilon(\Delta NX) = (\Delta\varepsilon)X + \varepsilon(\Delta X) - (\Delta IM) \quad (18A.2)$$

Derivation of the Marshall-Lerner Condition (3)

18.4 Depreciation, the Trade Balance, and Output

- Divide both sides of (18A.2) by εX to get:

$$\frac{\varepsilon(\Delta NX)}{\varepsilon X} = \frac{(\Delta \varepsilon)X}{\varepsilon X} + \frac{\varepsilon(\Delta X)}{\varepsilon X} - \frac{\Delta IM}{\varepsilon X}$$

- Simplify, and use the fact that, if trade is initially balanced, $\varepsilon X = IM$ to replace εX by IM in the last term on the right. This gives

$$\frac{\Delta NX}{X} = \frac{\Delta \varepsilon}{\varepsilon} + \frac{\Delta X}{X} - \frac{\Delta IM}{IM} \quad (18A.3)$$

Derivation of the Marshall-Lerner Condition (4)

18.4 Depreciation, the Trade Balance, and Output

- The result is intuitive. The change in the trade balance (as a ratio to exports) in response to a real depreciation is the sum of three items:
 - ① The first term is equal to the proportional change in the real exchange rate. It is negative if there is a real depreciation.
 - ② The second term is equal to the proportional change in exports. It is positive if there is a real depreciation.
 - ③ The third term is equal to minus the proportional change in imports. It is positive if there is a real depreciation.
- The Marshall-Lerner condition is the condition that the sum of these three terms be positive. If it is satisfied, a real depreciation leads to an improvement in the trade balance.

A Numerical Example

18.4 Depreciation, the Trade Balance, and Output

- Suppose that a 1% depreciation leads to a proportional increase in exports of 0.9% and a proportional decrease in imports of 0.8%.
- (Econometric evidence on the relation of exports and imports to the real exchange rate suggests that these are reasonable numbers.)
- In this case, the right-hand side of (18A.3) is equal to $-1\% + 0.9\% - (-0.8)\% = 0.7\%$.
- Thus, the trade balance improves, and the Marshall-Lerner condition is satisfied.

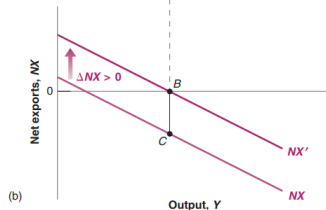
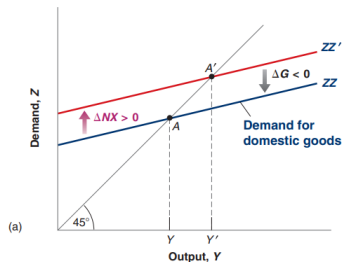
Combining Exchange Rate and Fiscal Policies

18.4 Depreciation, the Trade Balance, and Output

- Suppose output is at its natural level but the economy is running a large trade deficit.
- The government would like to reduce the trade deficit while leaving output unchanged so as to avoid overheating. What should it do?
- A depreciation alone will not do. Nor will a fiscal contraction do. Use the right combination of depreciation and fiscal contraction.

Reducing the Trade Deficit without Changing Output

18.4 Depreciation, the Trade Balance, and Output



Exchange Rate and Fiscal Policy Combinations

18.4 Depreciation, the Trade Balance, and Output

- To the extent that governments care about *both* the level of output and the trade balance, they have to use *both* fiscal policy and exchange rate policies.

Table 18-1 Exchange Rate and Fiscal Policy Combinations		
Initial Conditions	Trade Surplus	Trade Deficit
Low output	$\epsilon? G\uparrow$	$\epsilon\downarrow G?$
High output	$\epsilon\uparrow G?$	$\epsilon? G\downarrow$

Saving and Investment in the Open Economy

18.5 Saving, Investment, and the Current Account Balance

- We have seen in Chapter 3 that we could rewrite the condition for equilibrium in the goods market as the condition that investment is equal to saving.
- We can now derive the corresponding condition for the open economy.
- Start from our equilibrium condition

$$Y = C + I + G + X - \frac{IM}{\varepsilon}$$

- Move consumption C from the right side to the left side of the equation, subtract taxes T , from both sides, and denote net exports $X - IM/\varepsilon$ by NX to get

$$Y - C - T = I + (G - T) + NX$$

The Current Account Balance

18.5 Saving, Investment, and the Current Account Balance

- Add net income from abroad NI to the both sides of the equation:

$$(Y + NI - T) - C = I + (G - T) + (NX + NI)$$

- Note that the term in parentheses on the left side is equal to disposable income, so the left side is equal to disposable income minus consumption (i.e., private saving S).
- Recall that the sum of net exports and net income from abroad on the right side is equal to the current account.
- Denote the current account by CA and rewrite the previous equation as:

$$S = I + (G - T) + CA$$

- Reorganize the equation to read:

$$CA = S + (T - G) - I \quad (18.5)$$

Interpretation of Equation (18.5)

18.5 Saving, Investment, and the Current Account Balance

Note some of the things that equation (18.5) says:

- An increase in investment must be reflected in either in increase in private or public saving or a deterioration of the current account balance—a smaller current account surplus, or a larger current account deficit, depending on whether the current account is initially in surplus or in deficit.
- A deterioration in the government budget balance—either a smaller budget surplus or a larger budget deficit—must be reflected either in an increase in private saving or a decrease in investment, or in a deterioration of the current account balance.
- A country with a high saving rate (private and public) must have either a high investment or a large current account surplus.